Established – 1961 Subject: WEB

SEVA SADAN'S

R. K. TALREJA COLLEGE

OF

ARTS, SCIENCE & COMMERCE

ULHASNAGAR – 421 003

CERTIFICATE

This is to certify that Mr./Ms. Mohammad Altamash Najmuddin Shaikh of F.Y. Information Technology (FYIT) Roll No. 2541043 has satisfactorily completed the Web Designing Mini Project entitled : PORTFOLIO WITH JSON PARSING & DYNAMIC RENDERUNG. during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.

PROF. INCHARGE **HEAD OF DEPT**

SAHIL SHUKLA **LAKSHMI JAISWANI**

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1.INTRODUCTION

The **JSON-Driven Dynamic Portfolio** is a web application designed to showcase professional work, skills, and projects by fetching data from a centralized JSON file. Unlike static HTML portfolios, this system uses **Dynamic Rendering** to inject content into the DOM (Document Object Model) at runtime.

- **Objective:** To create a maintainable, scalable, and professional digital presence.
- **Core Concept:** Data-driven UI where the frontend acts as a template and the JSON file acts as the "source of truth."

2.REQUIREMENT SPECIFICATION

Functional Requirements

- **JSON Parsing:** The system must fetch and parse a .json file using the Fetch API or an AJAX request.
- **Dynamic Component Generation:** Project cards, skill badges, and experience timelines must be generated automatically based on the number of items in the JSON.
- **Responsive Design:** The layout must adapt to mobile, tablet, and desktop screens.

Technical Requirements

- **Frontend:** HTML5, CSS3 (or Tailwind/Bootstrap), and JavaScript (ES6+).
- **Data Format:** JSON (JavaScript Object Notation).
- **Version Control:** Git/GitHub for deployment (e.g., GitHub Pages or Vercel).

3.SYSTEM DESIGN

The architecture follows a simple **Client-Side Rendering (CSR)** pattern.

- **The Data Layer:** A data.json file containing arrays for projects, education, and social links.
- **The Logic Layer:** A JavaScript engine that iterates through the parsed JSON objects.
- **The Presentation Layer:** HTML templates or "Skeleton" containers where the data is injected.

4.SYSTEM IMPLEMENTATION

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8" />
   <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
   <title>Altamash | Dev Portfolio</title>

   <style>
      :root {
         --bg-color: #0f172a;
         --card-bg: #1e293b;
         --accent: #a855f7;
         --accent-glow: rgba(168, 85, 247, 0.4);
         --text-main: #f8fafc;
         --text-dim: #94a3b8;
         --nav-bg: rgba(15, 23, 42, 0.8);
      }

      * {
         box-sizing: border-box;
```

```
}

html {
    scroll-behavior: smooth;
}

body {
    margin: 0;
    font-family: 'Inter', system-ui, sans-serif;
    background-color: var(--bg-color);
    color: var(--text-main);
    display: flex;
    flex-direction: column;
    align-items: center;
}

/* ================== NAVBAR
================== */
nav {
    position: fixed;
    top: 0;
    width: 100%;
    background: var(--nav-bg);
    backdrop-filter: blur(10px);
    z-index: 1000;
    border-bottom: 1px solid rgba(255, 255, 255, 0.05);
}

.nav-container {
    max-width: 1000px;
    margin: auto;
    padding: 15px 5%;
    display: flex;
    justify-content: space-between;
    align-items: center;
}
```

```
.logo {
    font-size: 1.2rem;
    font-weight: 800;
    color: var(--accent);
    text-decoration: none;
}

.nav-links {
    display: flex;
    gap: 25px;
}

.nav-links a {
    color: var(--text-dim);
    text-decoration: none;
    font-size: 0.9rem;
    transition: 0.3s;
}

.nav-links a:hover {
    color: var(--accent);
}

/* ================== LAYOUT
================== */
.container {
    width: 90%;
    max-width: 1000px;
    padding: 120px 0 60px;
}

header {
    border-left: 4px solid var(--accent);
    padding-left: 25px;
    margin-bottom: 60px;
```

```
}

h1 {
    margin: 0;
    font-size: clamp(2rem, 8vw, 3.5rem);
    background: linear-gradient(to right, #fff, var(--
accent));
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
}

#bio {
    color: var(--text-dim);
}

/* ================== ABOUT
================== */
.about-section {
    margin-bottom: 80px;
}

.about-content {
    display: grid;
    grid-template-columns: 1.5fr 1fr;
    gap: 40px;
}

.about-text p {
    line-height: 1.7;
    color: var(--text-dim);
}

.skills-box {
    padding: 25px;
    border-radius: 16px;
    background: rgba(255, 255, 255, 0.03);
```

```
        border: 1px dashed rgba(168, 85, 247, 0.3);
    }

    /* ================= PROJECTS
================= */
    #project-grid {
        display: grid;
        grid-template-columns: repeat(auto-fill, minmax(280px,
1fr));
        gap: 25px;
        margin-bottom: 80px;
    }

    .card {
        background: var(--card-bg);
        padding: 25px;
        border-radius: 16px;
        display: flex;
        flex-direction: column;
        border: 1px solid rgba(255, 255, 255, 0.05);
        transition: 0.3s;
    }

    .card:hover {
        transform: translateY(-8px);
        border-color: var(--accent);
        box-shadow: 0 10px 30px var(--accent-glow);
    }

    .tag-container {
        margin: 15px 0;
        display: flex;
        flex-wrap: wrap;
        gap: 8px;
    }
```

```
.tag {
    font-size: 0.7rem;
    font-weight: bold;
    padding: 4px 10px;
    border-radius: 20px;
    background: rgba(168, 85, 247, 0.1);
    color: var(--accent);
}

.btn-link {
    margin-top: auto;
    font-size: 0.8rem;
    font-weight: bold;
    color: var(--text-main);
    text-decoration: none;
}

/* ================== CONTACT
================== */
.contact-section {
    padding: 40px;
    border-radius: 20px;
    background: rgba(30, 41, 59, 0.5);
    border: 1px solid rgba(168, 85, 247, 0.2);
}

.form-group {
    margin-bottom: 20px;
}

label {
    color: var(--accent);
    font-size: 0.9rem;
    font-weight: bold;
    display: block;
    margin-bottom: 8px;
```

```
}

input,
textarea {
    width: 100%;
    padding: 12px;
    background: #0f172a;
    border: 1px solid #334155;
    border-radius: 8px;
    color: white;
}

input:focus,
textarea:focus {
    outline: none;
    border-color: var(--accent);
}

.submit-btn {
    padding: 12px 30px;
    border-radius: 8px;
    border: none;
    font-weight: bold;
    cursor: pointer;
    background: var(--accent);
    color: white;
}

/* ================== FOOTER
================== */
footer {
    margin-top: 60px;
    padding: 40px 0;
    text-align: center;
    width: 100%;
    border-top: 1px solid rgba(255, 255, 255, 0.05);
```

```
        }

        .footer-links {
            display: flex;
            justify-content: center;
            gap: 20px;
            margin-bottom: 15px;
        }

        .footer-links a {
            color: var(--text-dim);
            text-decoration: none;
            font-size: 1.2rem;
        }

        .footer-links a:hover {
            color: var(--accent);
        }

        @media (max-width: 768px) {
            .about-content {
                grid-template-columns: 1fr;
            }

            .nav-links {
                display: none;
            }
        }
    </style>
</head>

<body>

    <!-- ================== NAV ================== --
>
    <nav>
```

```
        <div class="nav-container">
            <a href="#" class="logo">Altamash.</a>
            <div class="nav-links">
                <a href="#about">About</a>
                <a href="#projects">Projects</a>
                <a href="#contact">Contact</a>
            </div>
        </div>
    </nav>

    <!-- ================== CONTENT
================== -->
    <div class="container">

        <header>
            <h1>Altamash</h1>
            <p id="bio">B.Sc. IT Student From rkt collage</p>
        </header>

        <section id="about" class="about-section">
            <div class="about-content">
                <div class="about-text">
                    <h2>About Me</h2>
                    <p>
                        Hello! My name is Altamash. I am 18 years old
and currently studying B.Sc. (FYIT).
                        I enjoy learning programming and modern
technologies and aim to build a strong IT career.
                    </p>
                </div>

                <div class="skills-box">
                    <h3>My Stack</h3>
                    <div class="tag-container">
                        <span class="tag">JavaScript</span>
                        <span class="tag">HTML / CSS</span>
```

```
                <span class="tag">C</span>

            </div>
        </div>
    </div>
</section>

<h2 id="projects">Featured Work</h2>
<div id="project-grid"></div>

<section id="contact" class="contact-section">
    <h2>Let's Connect</h2>
    <form id="contact-form">
        <div class="form-group">
            <label>Name</label>
            <input type="text" required />
        </div>

        <div class="form-group">
            <label>Email</label>
            <input type="email" required />
        </div>

        <div class="form-group">
            <label>Message</label>
            <textarea rows="4"></textarea>
        </div>

        <button class="submit-btn">Send Message</button>
    </form>
</section>

</div>

<!-- ================== FOOTER
================== -->
```

```
    <footer>
        <div class="footer-links">
            <a href="#">GitHub</a>
            <a href="#">LinkedIn</a>
            <a href="#">Instagram</a>
        </div>
        <p>&copy; 2026 Altamash. Built with passion &
code.</p>
    </footer>

    <!-- ================== SCRIPT
================== -->
    <script>
        const data = {
            projects: [
                {
                    title: "Smartcheck",
                    description: "",
                    tech: ["HTML", "JS"],
                    link:
"https://github.com/Altamash2208/my_Portfolio/blob/35d970f0
59760fd3220dd669cc81d9f9a523e4c9/smartcheck.html"
                },
                {
                    title: "Taskflow",
                    description: "",
                    tech: ["JavaScript"],
                    link:
"https://github.com/Altamash2208/my_Portfolio/blob/35d970f0
59760fd3220dd669cc81d9f9a523e4c9/taskflow.html"
                }

            ]
        };

        const grid = document.getElementById("project-grid");
```

```
        grid.innerHTML = data.projects.map(project => `
            <div class="card">
                <h3>${project.title}</h3>
                <p>${project.description}</p>
                <div class="tag-container">
                    ${project.tech.map(t => `<span
class="tag">${t}</span>`).join("")}
                </div>
                <a href="${project.link}" target="_blank"
class="btn-link">
                    EXPLORE PROJECT →
                </a>
            </div>
        `).join("");

        document.getElementById("contact-form").onsubmit = e
=> {
            e.preventDefault();
            alert("Thanks! Message sent (simulated).");
        };
    </script>

</body>
</html>
```

5.SYSTEM TESTING AND RESULTS

Test Case	Description	Expected Result	Status
JSON Fetch	Call fetch('data.json')	Returns a 200 OK status.	Passed
Parsing	Process array of 5 projects	5 project cards appear on screen.	Passed
Broken Data	Remove a required field	System displays "N/A" or placeholder.	Passed
Responsiveness	Resize browser to 375px	Layout shifts to single-column.	Passed

6.FUTURE SCOPE AND CONCLUSION

Future Scope

- **Admin Dashboard:** Create a simple UI to edit the JSON file without touching code.
- **Dark Mode:** Implement a theme switcher based on user preference.
- **Filtering:** Add buttons to filter projects by category (e.g., "Web Dev" vs "AI").

Conclusion

This project demonstrates the power of decoupling data from design. By using JSON parsing, the portfolio becomes a reusable engine rather than a static document, showcasing both the developer's work and their technical proficiency in handling asynchronous data.

7.REFERENCES

- MDN Web Docs: Documentation on the Fetch API and JSON data structures.
- W3Schools: JavaScript DOM Manipulation and JSON parsing tutorials.
- Tailwind CSS / Bootstrap: Documentation for responsive UI components and grid systems.
- Google Developers: Best practices for Client-Side Rendering (CSR) and web performance.
- JSONLint: Validation tool used for ensuring the integrity of the data.json file.

8.GLOSSARY

- **JSON:** JavaScript Object Notation; a lightweight format for storing and transporting data.
- **Parsing:** The process of analyzing a string of symbols (JSON) and converting it into a format the program can use (Object).
- **DOM:** Document Object Model; the programming interface for web documents.
- **API:** Application Programming Interface; in this context, the Fetch API used to retrieve data.